

Ritik Vaishnav

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EDUCATION

- MBM University** Jodhpur, India
Bachelor of Engineering - Electronics and Electrical Engineering July 2021 - June 2026

EXPERIENCE

- Wireless Systems Engineering Lab - IIIT Delhi** Delhi, India
Research Intern under Dr. Arani Bhattacharya June 2025 - October 2025
 - Designed and implemented a stereo-guided adaptive super-resolution pipeline achieving 35% improvement in far-object detection ($>100\text{m}$) for autonomous vehicles.
 - Implemented StereoSGBM-based disparity estimation and developed depth-based region partitioning to identify far-field regions for selective super-resolution.
 - Developed a depth-aware selective tiling framework that reduces SR computation by $3\times$ compared to full-frame processing.
 - Optimized the end-to-end Stereo-SR pipeline on GPU (RTX 3060) for real-time performance ($\sim 80\text{ ms/frame}$).
 - Validated the framework on CARLA and KITTI, demonstrating 27% improvement in 50–100m detection with depth estimation MAE of 5.70m.
- Wireless Systems Engineering Lab - IIIT Delhi** Delhi, India
Research Intern under Dr. Arani Bhattacharya June 2024 - July 2024
 - Implemented efficient video encoding and streaming pipeline using FFmpeg (H.264/H.265) and RTSP in CARLA, achieving $>80\%$ bandwidth reduction with PSNR $>30\text{ dB}$.
 - Enabled real-time remote teleoperation through keyboard-based vehicle control integration in CARLA simulator.
 - Optimized Pylot perception pipeline by replacing Faster-RCNN with YOLOv8n, enabling dynamic cloud-based model switching to YOLOv8x for complex scenarios.
- Embedded Systems & Robotics Workshop** Jodhpur, India
Robotics Mentor June 2023 - August 2023
 - Mentored a batch of 100+ undergraduates in electronics and programming, sharing knowledge gained from ESRC under Prof. Alok Singh Gahlot.

PUBLICATIONS & RECOGNITION

- Depth-Aware Adaptive Video Streaming for Safe and Efficient Remote Autonomous Vehicle Supervision [Paper]**
Ritik Vaishnav and Arani Bhattacharya [ACM MMSys DS]
ACM Multimedia Systems Doctoral Symposium Apr 2026
- How Far Is Too Far? Fixing Vision of Autonomous Vehicles Using Selective Super-Resolution [Paper]**
Najiya Naj, Ritik Vaishnav, and Arani Bhattacharya [COMSNETS-W]
Workshop on Machine Intelligence in Networked Data and Systems Jan 2026
- Best Presentation Award** [RIISE]
Poster: “How Far is Too Far? Fixing AV vision with the cloud”
RIISE 2025, IIIT Delhi Sept 2025

PROJECTS

- Autonomous Rover for Vision-Based Target Detection and Navigation** April 2026
 - Built a low-cost autonomous rover running on ESP32 using a split-processing architecture, for object detection and autonomous navigation.
 - Implemented a 360° environment scan pipeline utilizing a mobile camera and YOLOv8, enabling autonomous target detection across varied indoor environments.
 - Developed end-to-end mission logic: scan $\rightarrow$ detect $\rightarrow$ align $\rightarrow$ approach, with bounding-box-based proximity estimation and real-time 2D dead-reckoning map using OpenCV.

- **StereoSR for Autonomous Vehicle Perception**

Sept 2025

- Implemented stereo vision pipeline for depth estimation using dual camera setup in CARLA simulator.
- Achieved strong accuracy with ground truth validation against CARLA's native depth sensor.
- Benchmarked the setup on KITTI Stereo dataset with an MAE of 5.70m, improving far-object detection by 35% beyond 100m.

POSITION OF RESPONSIBILITY

- **Founder, Pixellens: Film and Photography Club**

Jodhpur, India

Founded and built the club from ground up

March 2022 - March 2025

- Grew it to a community of 50+ super creative people
- Created and led multiple creative film and photography projects
- Written/Directed/Shot/Edited short films in 50Hr timeline - 3x

SKILLS

- **Languages:** Python
- **Frameworks:** PyTorch, OpenCV, YOLO
- **Tools:** FFmpeg, CARLA, Pylot, Git, Linux

INTERESTS

- **Research Interests:** Autonomous Vehicle Perception, Real-time Systems, Computer Vision, Hardware-Software Co-design
- **Abstract Interests:** Filmmaking, Music